

Supplementary Material for Variance-Corrected Multi-Asset Equity Simulation with Hybrid Hidden Markov Marginals

S1. Empirical cross-term diagnostic

The variance correction of Eq. (8) assumes $\text{Cov}(\tilde{g}_i^c, g_m) \approx 0$, allowing the variance derivation to omit the cross-term. This assumption does not follow automatically from the fitting procedure because each JumpHMM uses the asset’s full growth-rate history, including its market-correlated component. A remaining market loading in the generator draw would invalidate the closed-form scale. We tested the assumption by sampling $R = 100$ generator growth-rate paths $\tilde{g}_i^{(r)}$ per ticker from each generator and computing the normalized cross-covariance against the real SPY path:

$$\text{cov}_i^{\text{norm}} = \frac{1}{R} \sum_{r=1}^R \frac{\text{Cov}((\tilde{g}_i^{(r)})^c, g_m)}{\sigma_{\text{gen},i} \sigma_m}. \quad (\text{S1})$$

Centering does not change covariance, so these values are numerically identical to those obtained from the raw draws. Per-ticker values were plotted as histograms in Supplementary Fig. S1. Across the 423 tickers, the full-return JumpHMM generator (used by the hybrid and naive composition methods), the residual-fit JumpHMM generator (used by the JumpHMM-on-residuals composition method), and the Gaussian baseline had universe-mean normalized cross-covariances of 1.53×10^{-4} , 6.58×10^{-5} , and 2.26×10^{-4} , respectively. Their per-ticker values ranged from -0.0061 to 0.0061 . These values quantify the cross-term omitted in Eq. (8).

S2. Bias correction for daily-rebalanced backtests

This appendix derives the bias correction in Eq. (S14). Under daily rebalancing, the raw synthetic ensemble had a median annualized growth rate about 22 percentage points above the long-run prior. The variance correction in Eq. (8) first sets each synthetic asset’s second moment to its generator target. When that target matches the empirical second moment, the second-order formula for the diversification return is also matched at the asset level. The remaining difference can reflect omitted cross-asset residual covariance, higher-order Taylor terms, calibration drift, and the documented attenuation of realized diversification

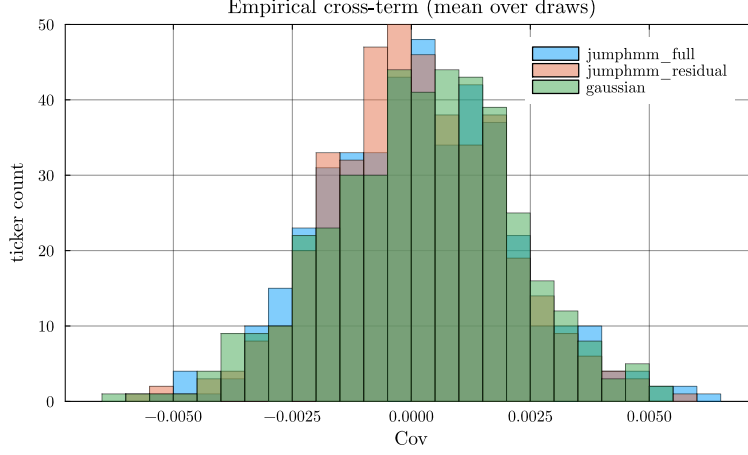


Figure S1: **Empirical normalized cross-covariance** $\text{Cov}(\tilde{g}_i^c, g_m)/(\sigma_{\text{gen},i} \sigma_m)$ **across the 423-ticker universe.** Histograms were stacked per generator, with each per-ticker value averaged over $R = 100$ generator growth-rate paths. Universe means were 1.53×10^{-4} , 6.58×10^{-5} , and 2.26×10^{-4} for the full-return JumpHMM, residual-fit JumpHMM, and Gaussian generators.

returns (Bouchey et al., 2015). The location-shift correction absorbs these effects as a single empirical drag on log-wealth.

The synthetic paths reproduce per-asset marginals and single-factor cross-sectional dependence. Each residual generator retains its own temporal structure, but the residual paths are generated independently across assets at each date. The construction therefore omits cross-asset residual covariance. Consider a daily-rebalanced portfolio of N assets with constant target weights $w_i \geq 0$, $\sum_i w_i = 1$, reset to those weights at the start of each trading day. Throughout this derivation $g_i(t)$ is the annualized log-growth rate of Eq. (1), so the per-period log return on asset i is $g_i(t) \Delta t$ with $\Delta t = 1/252$ yr. The wealth process is:

$$W_t = W_{t-1} \sum_i w_i \exp(g_i(t) \Delta t), \quad (\text{S2})$$

so the per-step log return on the portfolio is:

$$\log\left(\frac{W_t}{W_{t-1}}\right) = \log\left[\sum_i w_i \exp(g_i(t) \Delta t)\right].$$

We suppress the time argument, writing g_i for $g_i(t)$. A second-order Taylor expansion of the exponential about zero gives:

$$\exp(g_i \Delta t) = 1 + g_i \Delta t + \frac{1}{2}(g_i \Delta t)^2 + O((g \Delta t)^3).$$

Substituting this expansion into each summand gives:

$$\begin{aligned}
\sum_i w_i \exp(g_i \Delta t) &= \sum_i w_i \left[1 + g_i \Delta t + \frac{1}{2} (g_i \Delta t)^2 \right. \\
&\quad \left. + O((g \Delta t)^3) \right] \\
&= \sum_i w_i + \Delta t \sum_i w_i g_i + \frac{1}{2} (\Delta t)^2 \sum_i w_i g_i^2 + O((g \Delta t)^3) \\
&= 1 + \Delta t \sum_i w_i g_i + \frac{1}{2} (\Delta t)^2 \sum_i w_i g_i^2 + O((g \Delta t)^3). \quad (\text{S3})
\end{aligned}$$

A second-order Taylor expansion of $\log(1+x)$ about $x=0$ gives $\log(1+x) = x - \frac{1}{2}x^2 + O(x^3)$. Identifying $1+x$ with the right-hand side of Eq. (S3), the order- Δt correction is $x := \Delta t \sum_i w_i g_i + \frac{1}{2} (\Delta t)^2 \sum_i w_i g_i^2$, and squaring keeps only the leading $(\Delta t)^2$ term: $x^2 = (\Delta t)^2 (\sum_i w_i g_i)^2 + O((\Delta t)^3)$ because the cross term has an extra Δt factor. Substitution gives:

$$\begin{aligned}
\log \frac{W_t}{W_{t-1}} &= \log \left(1 + x + O((g \Delta t)^3) \right) \\
&= x - \frac{1}{2} x^2 + O(x^3) \\
&= \Delta t \sum_i w_i g_i + \frac{1}{2} (\Delta t)^2 \sum_i w_i g_i^2 \\
&\quad - \frac{1}{2} (\Delta t)^2 \left(\sum_i w_i g_i \right)^2 + O((g \Delta t)^3) \\
&= \Delta t \sum_i w_i g_i + \frac{1}{2} (\Delta t)^2 \left[\sum_i w_i g_i^2 \right. \\
&\quad \left. - \left(\sum_i w_i g_i \right)^2 \right] + O((g \Delta t)^3). \quad (\text{S4})
\end{aligned}$$

Dividing through by Δt extracts the portfolio's annualized one-step log-growth rate $g_p(t) := \Delta t^{-1} \log(W_t/W_{t-1})$, the portfolio counterpart of the per-asset $g_i(t)$ (still suppressing the time argument):

$$g_p = \sum_i w_i g_i + \frac{1}{2} \Delta t \left(\sum_i w_i g_i^2 - \left(\sum_i w_i g_i \right)^2 \right) + O((\Delta t)^2). \quad (\text{S5})$$

Restoring the time argument, we now take unconditional expectations of Eq. (S5) under stationarity. Define the per-asset annualized mean and variance:

$$\mu_i := \mathbb{E}(g_i(t)), \quad \sigma_i^2 := \text{Var}(g_i(t)), \quad (\text{S6})$$

the per-rate covariance $\Sigma_{ij} := \text{Cov}(g_i(t), g_j(t))$, and the per-rate portfolio variance $\sigma_p^2 := \text{Var}(\sum_i w_i g_i(t)) = w^\top \Sigma w$. The $O(\Delta t)$ correction in Eq. (S5) contains two squared quantities, $\sum_i w_i g_i(t)^2$ and $(\sum_i w_i g_i(t))^2$. Both expand

by the second-moment identity, which holds for any random variable X with finite second moment:

$$\mathbb{E}(X^2) = \mathbb{E}(X)^2 + \text{Var}(X). \quad (\text{S7})$$

Applying this identity to $X = g_i(t)$ and summing against w_i gives:

$$\begin{aligned} \mathbb{E}\left(\sum_i w_i g_i(t)^2\right) &= \sum_i w_i \mathbb{E}(g_i(t)^2) && (\text{linearity}) \\ &= \sum_i w_i [\mu_i^2 + \sigma_i^2] && (\text{by Eq. (S7)}) \\ &= \sum_i w_i \mu_i^2 + \sum_i w_i \sigma_i^2. \end{aligned} \quad (\text{S8})$$

Applying the identity to $X = \sum_i w_i g_i(t)$, whose mean is $\sum_i w_i \mu_i$ and whose variance is σ_p^2 , gives:

$$\begin{aligned} \mathbb{E}\left(\left(\sum_i w_i g_i(t)\right)^2\right) &= \mathbb{E}\left(\sum_i w_i g_i(t)\right)^2 + \text{Var}\left(\sum_i w_i g_i(t)\right) \quad (\text{by Eq. (S7)}) \\ &= \left(\sum_i w_i \mu_i\right)^2 + \sigma_p^2. \end{aligned} \quad (\text{S9})$$

We substituted Eqs. (S8)–(S9) into the expectation of Eq. (S5) and grouped the variance and mean contributions as follows:

$$\begin{aligned} \mathbb{E}(g_p) &= \sum_i w_i \mu_i + \underbrace{\frac{1}{2} \Delta t \left(\sum_i w_i \sigma_i^2 - \sigma_p^2 \right)}_D \\ &\quad + \frac{1}{2} \Delta t \left[\sum_i w_i \mu_i^2 - \left(\sum_i w_i \mu_i \right)^2 \right] + O((\Delta t)^2), \end{aligned} \quad (\text{S10})$$

the Booth-Fama identity for the daily-rebalanced portfolio in annualized growth-rate units (Booth and Fama, 1992; Markowitz, 1976). The middle term, D , is the diversification return: the $O(\Delta t)$ correction equal to half the difference between the weighted-average per-asset variance $\sum_i w_i \sigma_i^2$ and the portfolio variance σ_p^2 . The per-rate moments μ_i and σ_i^2 have units yr^{-1} and yr^{-2} respectively; the products $\Delta t \sigma_i^2$ and $\Delta t \sigma_p^2$ in Eq. (S10) are the annualized variances (yr^{-1}). The third term in Eq. (S10) is the cross-sectional dispersion of the per-asset annualized means, of order $\Delta t \mu^2 \sim 3 \times 10^{-5} \text{ yr}^{-1}$ at $\mu \sim 0.08 \text{ yr}^{-1}$ and negligible against the variance term $D \sim \Delta t \sigma^2 \sim 4 \times 10^{-2} \text{ yr}^{-1}$; we drop it. The realized annualized compound growth rate $\bar{g}_p = (T \Delta t)^{-1} \log(W_T/W_0) = T^{-1} \sum_t g_p(t)$ converges to $\mathbb{E}(g_p)$ by the ergodic theorem under any stationary residual process, so the long-run limit is fixed by the one-time-slice unconditional joint distribution of $(g_i(t))_i$ and does not, in itself, depend on the serial structure of the residual draw.

To rewrite D in a form that makes its non-negativity manifest, expand $\sigma_p^2 = \sum_i w_i^2 \sigma_i^2 + 2 \sum_{i < j} w_i w_j \Sigma_{ij}$ and use the identity $\sum_i w_i(1 - w_i) \sigma_i^2 = \sum_{i < j} w_i w_j (\sigma_i^2 + \sigma_j^2)$ that holds for any weights summing to one:

$$D = \frac{1}{2} \Delta t \left(\sum_i w_i \sigma_i^2 - \sigma_p^2 \right) = \frac{1}{2} \Delta t \sum_{i < j} w_i w_j \text{Var}(g_i - g_j) \geq 0, \quad (\text{S11})$$

so D equals one-half the annualized weighted-average pairwise growth-rate-difference variance and is therefore non-negative, with equality only when every pair (i, j) comoves perfectly ($\text{Var}(g_i - g_j) = 0$) or all weight concentrates on a single asset. The rebalancing rule itself produces this effect: weights drift between rebalances, the rebalancer sells whatever has outperformed (its weight rose above target) and buys whatever has underperformed (its weight fell below target), and the trade systematically converts period-by-period cross-sectional dispersion into compounded growth. Markowitz (1976) described this as the variance bonus of the long-run mean-variance allocation; Booth and Fama (1992) called it the diversification return.

Substituting the single-index decomposition $g_i(t) = \alpha_i + \beta_i g_m(t) + \varepsilon_i(t)$, with market variance $\sigma_m^2 := \text{Var}(g_m(t))$, per-asset residual variance $\sigma_{\varepsilon,i}^2 := \text{Var}(\varepsilon_i(t))$, and $\text{Cov}(g_m, \varepsilon_i) = 0$ (verified in Sec. S1), into Eq. (S11) gives:

$$\text{Var}(g_i - g_j) = (\beta_i - \beta_j)^2 \sigma_m^2 + \text{Var}(\varepsilon_i - \varepsilon_j), \quad (\text{S12})$$

Because g_m and the residuals are uncorrelated, this relation splits the diversification return into market-beta-dispersion and idiosyncratic components:

$$D = \underbrace{\frac{1}{2} \Delta t \sigma_m^2 \sum_{i < j} w_i w_j (\beta_i - \beta_j)^2}_{D_{\text{mkt}}} + \underbrace{\frac{1}{2} \Delta t \sum_{i < j} w_i w_j \text{Var}(\varepsilon_i - \varepsilon_j)}_{D_{\varepsilon}}, \quad (\text{S13})$$

where the market-beta-dispersion component D_{mkt} equals $\frac{1}{2} \Delta t \sigma_m^2 [\sum_i w_i \beta_i^2 - (\sum_i w_i \beta_i)^2]$, the weighted cross-sectional β dispersion scaled by the annualized market variance $\Delta t \sigma_m^2$. Across the 423-ticker universe the empirical β distribution spans $[0.03, 1.79]$ with interquartile range $[0.81, 1.18]$ and the SPY annualized variance $\Delta t \sigma_m^2 \approx 0.04 \text{ yr}^{-1}$. The weighted variance of β is bounded by one quarter of its squared range, so D_{mkt} is at most 0.016 yr^{-1} , or 1.6 percentage points per year over this universe. For the idiosyncratic component D_{ε} , residuals were approximately orthogonal across assets at the same time slice. Therefore, $\text{Var}(\varepsilon_i - \varepsilon_j) \approx \sigma_{\varepsilon,i}^2 + \sigma_{\varepsilon,j}^2$, and D_{ε} scales with the weighted-average annualized idiosyncratic variance $\Delta t \sigma_{\varepsilon,i}^2$ per name.

The second-order value of D depends on same-time unconditional moments, not on serial dependence. Equation (8) sets $\sigma_{\text{gen},i}^2 = \beta_i^2 \sigma_m^2 + \sigma_{\varepsilon,i}^2$ to its empirical target. Any residual generator with the same unconditional second moments therefore gives the same D . This includes a stationary AR(1) residual generator $\varepsilon_i(t) = \phi \varepsilon_i(t-1) + \eta_i(t)$ with i.i.d. innovations η_i tuned so that $\text{Var}(\varepsilon_i) = \sigma_{\eta,i}^2 / (1 - \phi^2) = \sigma_{\varepsilon,i}^2$. The parameter ϕ leaves the marginal distribution at a single

Table S1: **Descriptive statistics and stylized-facts tests for SPY daily excess growth rates.** In-sample (IS): 2014–2024 ($T = 2,766$); out-of-sample (OoS): 2025 ($T = 249$). Gaussian and Laplace columns show maximum likelihood estimation (MLE) fits to the in-sample data. We used the Jarque–Bera (JB) normality test and the Ljung–Box (LB) autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed	Gaussian	Laplace
Mean (annualized, %)	6.31	11.60	6.31	14.19
Std. dev. of G_t (% yr ⁻¹)	214.50	220.62	214.46	205.93
Skewness	-0.753	-1.093	0	0
Excess Kurtosis	7.715	6.867	0	3
JB normality test	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a
LB test on G_t (lag 20)	reject* ($p < 0.001$)	reject* ($p = 0.019$)	n/a	n/a
LB test on $ G_t $ (lag 20)	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a

time slice unchanged. It therefore does not change $\mathbb{E}(g_p)$ in Eqs. (S10) and (S13), but it changes sampling variance. For a stationary AR(1) residual with variance σ_ε^2 , the variance of its sample mean is approximately $T^{-1}\sigma_\varepsilon^2(1+\phi)/(1-\phi)$. Under ergodicity, serial dependence changes per-path sampling noise, not the stationary mean. The raw median’s ~ 22 -percentage-point excess over an external long-run growth prior is therefore not caused by i.i.d. sampling *across time* at the level of the second-order formula.

The direct structural difference is contemporaneous. The derivation assumes residuals are orthogonal across assets at each time slice, as they are in the independent per-asset generator. Real residuals share sector and style factors, which reduce $\text{Var}(\varepsilon_i - \varepsilon_j)$ below the independent-generator value $\sigma_{\varepsilon,i}^2 + \sigma_{\varepsilon,j}^2$. Matching each marginal variance does not restore this covariance. The synthetic D_ε will therefore exceed its real-market counterpart when the omitted covariance is positive. Measuring that contribution requires a joint-residual experiment, which we did not run. The gap may also include higher-moment corrections to the Taylor expansion (heavy-tailed JumpHMM marginals have non-trivial third and fourth cumulants), the time-varying volatility and correlation across the 2014–2024 calibration window relative to a long-run prior, and the documented attenuation of realized diversification returns on real equity portfolios at all rebalancing horizons (Bouchey et al., 2015). The location-shift correction of Eq. (S14) treats this composite shortfall as a single empirical drag b to be absorbed and is agnostic to its decomposition.

To measure the bias, we applied a 20-asset, long-only target-weight allocator to the hybrid-composed universe. We calibrated the assets over 2014–2024, deployed the allocator in 2025, limited the universe to names with calibrated (α, β) and $\max \beta = 1.21$, and assumed a 5-basis-point round-trip cost. We then decomposed the path-mean annualized growth rate $\bar{g}_p \equiv (T \Delta t)^{-1} \log(W_{p,T}/W_{p,0})$ into static and rebalancing-driven components (Supplementary Table S2). The static-to-daily-rebalance switch increased the median by 27.73 percentage points. Adding a signal after daily rebalancing changed it by another 0.13 percentage points. The same allocator on the real 2025 growth-rate history produced 11.5%/yr, about 18 percentage points below the synthetic median. The synthetic residuals

Table S2: **Diversification-return decomposition on a 20-asset target-weight allocator deployed against the hybrid-composed 2025 universe.** Median path-mean annualized growth rate $\bar{g}_p$ across the synthetic ensemble. Equal-weight buy-and-hold serves as a passive baseline; the target-weight allocation is reported under static (no intra-year rebalance) and daily-rebalanced regimes, with and without a signal on top of the target weights. The static-to-daily increase of +27.73 pp on the same target weights isolates the diversification return of Eq. (S13). The same allocator deployed against the real 2025 growth-rate history is shown for comparison.

Allocation rule (synthetic 2025 universe)	$\bar{g}_p$ (%/yr)
Equal-weight buy-and-hold	7.68
Target-weight, static (no rebalance)	1.97
Target-weight, daily rebalanced	29.69
Target-weight, daily rebalanced + signal	29.83
<i>Real 2025 history, same allocator</i>	11.50

were independent across tickers and therefore omitted the contemporaneous covariance present in real markets. Determining how much of the gap this omission caused would require a joint-residual experiment with the marginals and allocator held fixed.

The correction subtracts a constant drift in log-wealth so that the ensemble median annualized growth rate matches an external prior $\bar{g}^{\text{prior}}$. At each time t , it multiplies every path by the same factor and therefore preserves cross-path wealth ranks. It also shifts every log-return increment by the same constant, which leaves the variance of those increments unchanged. For path p , we applied the following correction:

$$W_{p,t}^{\text{corr}} = W_{p,t} \cdot \exp(-b \cdot t \Delta t), \quad b = \underset{p}{\text{median}} \bar{g}_p - \bar{g}^{\text{prior}}, \quad (\text{S14})$$

with Δt the integration step in years and b , $\bar{g}_p$, $\bar{g}^{\text{prior}}$ all in the same per-year growth-rate units. The correction is a linear detrend in log-wealth and remains multiplicative in wealth. It changes drawdown dates and depths, so it adjusts the return level rather than preserving drawdowns. It does preserve terminal-wealth ranks and the variance of log-return increments. We set $\bar{g}^{\text{prior}} = 8\%/yr$ before examining the holdout result. This value was based on external long-run equity-return estimates and an allowance for fees, turnover, and market impact (Damodaran, 2025; Ibbotson and Duff and Phelps, 2024). We used this documented prior rather than the realized test year to avoid circular calibration. Matching the realized year would force the synthetic median to equal the test sample. In this deployment, the real-2025 outcome of 11.5%/yr fell just above the median of the corrected ensemble.

Several alternative fixes were considered and rejected. A path-filtering rule that drops synthetic paths above a $\bar{g}_p$ threshold turns the threshold itself into a tuning target and removes only a subset of the bias’s consequences while leaving the underlying bias intact. Inflating per-trade transaction costs to absorb the diversification return would require round-trip costs of ~ 70 basis

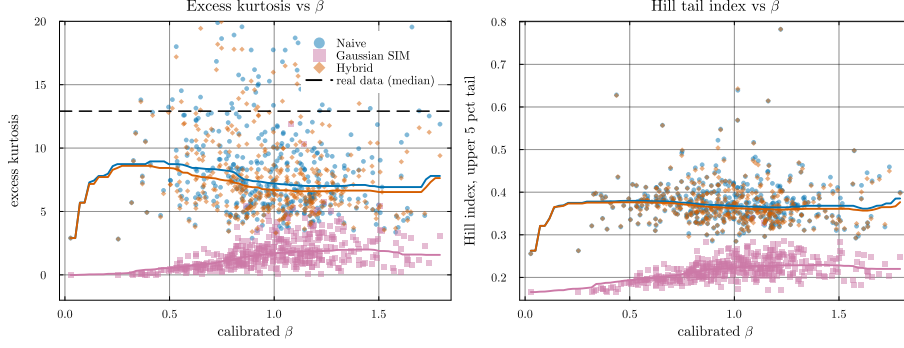


Figure S2: **Heavy-tail preservation: excess kurtosis and Hill tail index vs calibrated β .** Each point is a per-ticker median over 100 replications. *Left:* excess kurtosis, y -axis clipped to $[-2, 20]$ for readability. The dashed line marks the median real-data excess kurtosis across the universe ($\kappa_{\text{real}} \approx 13$); the hybrid and naive composition methods (orange and blue) track the generator's kurtosis, while Gaussian SIM collapses to $\kappa \approx 1$. *Right:* Hill tail-index estimator on the upper 5% of $|g|$. The hybrid and naive medians remain near 0.37 throughout the β range; Gaussian SIM drops to ~ 0.22 . The hybrid's median being below κ_{real} reflects the JumpHMM marginal's own kurtosis rather than a composition-rule distortion: hybrid tracks naive (both retain the generator's tails) while Gaussian SIM does not.

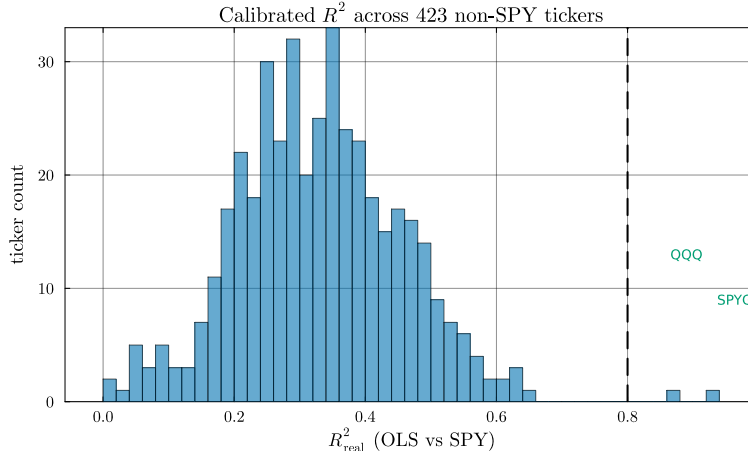


Figure S3: **Calibrated R^2 distribution across the 423 non-SPY tickers.** Histogram of OLS R^2 against SPY over the 2014–2024 window. The two tickers crossing the $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) are index ETFs (QQQ, SPYG); the highest- R^2 individual stock (BLK, 0.645) falls ~ 0.16 below the threshold. The R^2 -preserving branch therefore contains only QQQ and SPYG in this universe.

points versus the 1–10 basis-point range assumed for liquid US equity, shifting the misrepresentation from the synthetic generator to its cost model. Drifting the calibrated parameters between calibration and deployment to widen the predictive envelope, with drift scale varied across $\{0, 1, 2, 3, 5\}$ multiples of the OLS standard error, left the median Sharpe ratio essentially unchanged across the sweep, so standard-error-scale parameter drift did not absorb the bias. Imposing an AR(1) on the residual draw would conflict with the copula reorder step because reordering destroys temporal structure. Preserving both structures would require a block-ordered copula sampler and therefore a new residual model. The per-ticker block bootstrap (Politis and Romano, 1994) preserves each ticker’s temporal residual structure but not covariance across tickers. A joint bootstrap or residual copula would be required for that purpose. The block bootstrap also replaces the JumpHMM marginal rather than correcting it. A multiplicative haircut on the raw-versus-static deviation, $W^{\text{corr}} = W_{\text{static}} \cdot (W_{\text{raw}}/W_{\text{static}})^k$, hits the same target at k chosen to match the prior, but it changes the rank ordering of paths and inherits the static allocation’s drawdown pattern.

The location-shift correction preserves percentile ranks and the relative ranking of strategies evaluated on the same paths. It also preserves centered moments of terminal log growth, including variance, skewness, and kurtosis. Because it changes the drawdown distribution, we evaluated drawdowns on the uncorrected ensemble. The correction does not repair residual autocorrelation, absolute Sharpe under daily rebalancing, or sensitivity to rebalancing frequency. Those targets require a residual model that preserves both temporal and cross-sectional dependence.

S3. Validation metric definitions

The distributional and tail metrics reported in the main results were defined as follows. The two-sample Kolmogorov-Smirnov statistic between the empirical cumulative distribution functions (CDFs) of the observed and simulated growth-rate samples (G_{obs} of length n and G_{sim} of length m) is $D = \sup_x |F_n(x) - F_m(x)|$ (Kolmogorov, 1933; Smirnov, 1948); the test rejects when D exceeds the critical value at level $\alpha = 0.05$. The Anderson-Darling statistic places greater weight on the tails via the integral $A^2 = nm/(n+m) \int [F_n(x) - F_m(x)]^2 / [F(x)(1 - F(x))] dF(x)$, where F is the pooled empirical CDF (Anderson and Darling, 1952). The Wasserstein-1 distance between two empirical distributions of equal length is the average of sorted-quantile differences:

$$W_1 = \frac{1}{n} \sum_{i=1}^n |G_{\text{obs}}^{(i)} - G_{\text{sim}}^{(i)}|, \quad (\text{S15})$$

with $G^{(i)}$ the i th order statistic. The Hill upper-tail index is the classical order-statistic estimator:

$$\hat{\xi} = \frac{1}{k-1} \sum_{i=1}^{k-1} \log \left(\frac{X_{(i)}}{X_{(k)}} \right), \quad (\text{S16})$$

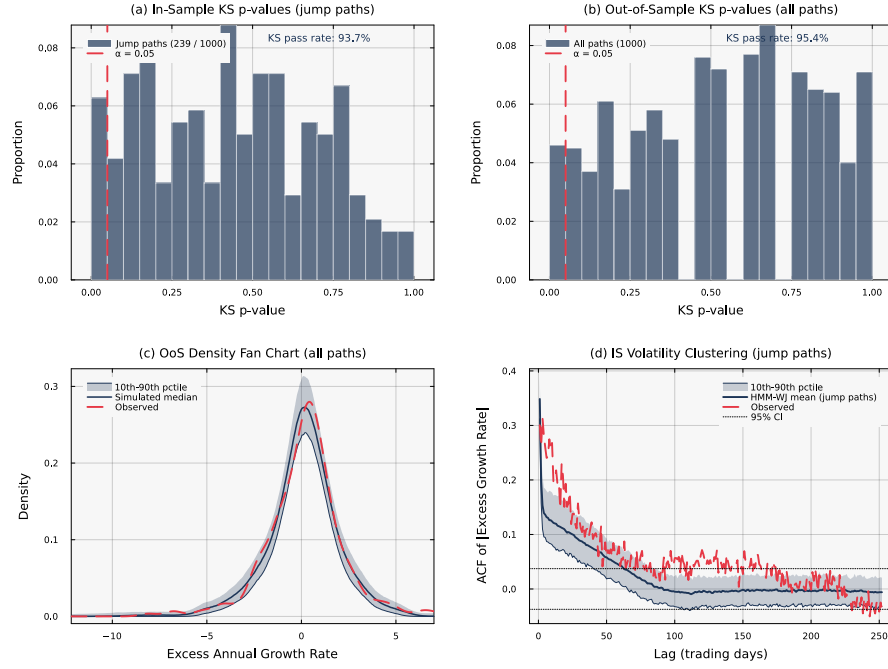


Figure S4: **Statistical validation of HMM-WJ** ($N = 100$, $\alpha = 0.05$). Panels (a) and (d) condition on the $\sim 24\%$ of in-sample paths containing jump events. (a) In-sample KS p -values for 239 jump paths (pass rate 93.7%). (b) Out-of-sample KS p -values for all 1,000 paths (pass rate 95.4%). (c) Out-of-sample density fan chart; the observed density falls within the simulation envelope. (d) Observed and mean simulated in-sample ACF of $|G_t|$ for jump paths.

with $X_{(1)} \geq X_{(2)} \geq \dots$ the ordered upper-tail absolute returns; k is set to the upper 5% of the sample. For a Pareto-tailed distribution with shape α , $\hat{\xi} \approx 1/\alpha$. For the Hellinger calculation, we binned both samples on a common grid and normalized the bin counts to probability masses p and q . Their distance is:

$$H(p, q) = \frac{1}{\sqrt{2}} \left\| \sqrt{p} - \sqrt{q} \right\|_2 \in [0, 1], \quad (\text{S17})$$

with $H = 0$ for identical probability masses and $H = 1$ for disjoint supports.

S4. Transition matrix internals and partition diagnostics

Supplementary Fig. S5 shows the fitted HMM-WJ components for SPY with $N = 100$ states. The fitted Laplace CDF and the empirical CDF are shown with the 99 state boundaries Q_k . The transition matrix $\mathbf{T}$ has a near-diagonal band, with most off-diagonal mass within ± 5 states of the diagonal. Under the ordinary Markov transitions, every state has an expected residence time of only 1–2 steps. The fitted jump duration is $\lambda = 90$ steps, about two orders of magnitude longer. Every state had at least one in-sample observation.

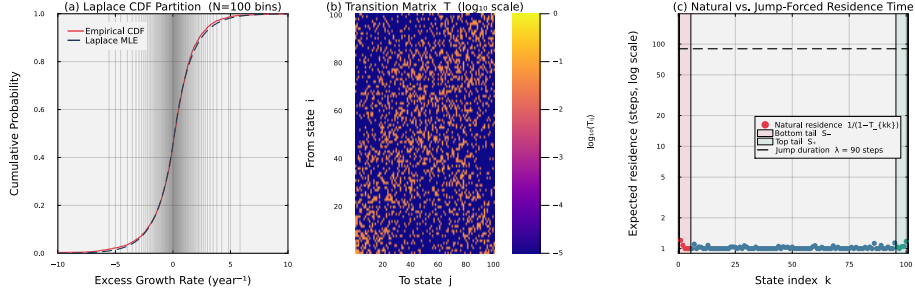


Figure S5: **Fitted HMM-WJ internals for SPY with $N = 100$ states.** Panel (a) overlays the fitted Laplace CDF on the empirical CDF; the 99 vertical lines mark the equal-probability quantile boundaries Q_k that define the hidden states. Panel (b) displays the estimated transition matrix $\mathbf{T}$ in $\log_{10}$ color scale; the dominant near-diagonal band reflects short-range regime persistence. Panel (c) plots the expected natural residence time $1/(1 - T_{kk})$ for each state on a $\log_{10}$ scale, with the bottom tail set $\mathcal{S}_-$ (states 1–5, red) and the top tail set $\mathcal{S}_+$ (states 96–100, teal) highlighted. The dashed horizontal line marks the fitted mean jump duration $\lambda = 90$ steps, quantifying the two-order-of-magnitude gap between natural residence times and the jump-duration override.

S5. Emission distribution and semi-Markov comparison

To compare the emission and duration specifications, we evaluated three variants on SPY (Supplementary Table S3): an HSMM-style baseline adapted from Bulla and Bulla (2006) ($K = 8$ states, negative-binomial dwell times with a geometric fallback, and Student- t emissions), HMM-WJ with the original Gaussian emissions ($N = 100$), and HMM-WJ with Student- t emissions ($N = 100$). Both HMM-WJ variants had higher in-sample KS and AD pass rates than the HSMM-style baseline. The HMM-WJ variants use 100 states and the HSMM uses 8, so this comparison does not assign the pass-rate difference solely to the duration model. Holding the rest of HMM-WJ fixed, changing the emission from Normal to Student- t ($\nu = 5$) increased simulated excess kurtosis from 5.5 to 7.5, against an observed value of 7.7. The Student- t emission also increased the KS and AD pass rates, while ACF-MAE and the distributional distances changed modestly.

S6. State-resolution sensitivity

We swept the partition resolution over $N \in \{30, 60, 90, 100, 150, 200, 350\}$ and re-evaluated the in-sample KS and AD pass rates for HMM-NJ and HMM-WJ. Pass rates varied little across $N \in \{60, \dots, 200\}$. At $N = 30$ both models lost several percentage points on the AD test. At $N = 350$ several quantile bins were never visited in the historical record, leaving the corresponding emission parameters undefined. We adopted $N = 100$ for all subsequent experiments.

Table S3: **Three-way comparison of emission distributions and duration specifications for SPY.** Standard errors in parentheses; bold marks the best value per row. The HSMM is a hidden semi-Markov-style model adapted from Bulla and Bulla (2006) ($K = 8$ states, negative-binomial dwell times with a geometric fallback, and Student- t emissions); HMM-WJ: hybrid hidden Markov model with jump-duration override ($N = 100$). All entries are computed from 1,000 simulated paths at significance level $\alpha = 0.05$.

Metric	HSMM (Student- t)	HMM-WJ (Gaussian)	HMM-WJ (Student- t)
States	$K = 8$	$N = 100$	$N = 100$
<i>In-sample (2,766 trading days, 2014-2024)</i>			
KS pass rate (%)	82.0 (1.2)	97.0 (0.5)	98.3 (0.4)
AD pass rate (%)	42.5 (1.6)	90.5 (0.9)	94.8 (0.7)
Excess kurtosis	4.8 (0.08)	5.5 (0.03)	7.5 (0.09)
ACF-MAE	0.059 (<0.001)	0.052 (<0.001)	0.053 (<0.001)
Wasserstein-1	0.176 (0.001)	0.101 (0.002)	0.097 (0.001)
Hellinger	0.113 (<0.001)	0.076 (<0.001)	0.074 (<0.001)
<i>Out-of-sample (249 trading days, 2025)</i>			
KS pass rate (%)	96.2 (0.6)	95.0 (0.7)	95.4 (0.7)
AD pass rate (%)	96.7 (0.6)	95.9 (0.6)	95.3 (0.7)
Excess kurtosis	4.1 (0.13)	4.9 (0.08)	6.2 (0.15)
ACF-MAE	0.042 (<0.001)	0.040 (<0.001)	0.040 (<0.001)
Wasserstein-1	0.287 (0.002)	0.275 (0.005)	0.275 (0.005)
Hellinger	0.239 (0.001)	0.212 (0.001)	0.210 (0.001)

Table S4: **Hybrid composition method broken out by branch.** N_{tickers} : number of tickers assigned to each branch by the $R^2_{\text{preserve}} = 0.80$ threshold. The clipped variance-preserving branch does not trigger anywhere in the universe ($\rho_i < 0.90$ for all tickers) and is omitted. The two trackers in the R^2 -preserving branch (QQQ and SPYG) had median absolute errors of 0.005 for β and 0.001 for R^2 , with a KS pass rate of 100%.

Branch	N_{tickers}	$ \hat{\beta} - \beta $	$ \hat{R}^2 - R^2_{\text{real}} $	KS pass (%)	W_1	κ
hybrid	421	0.018	0.021	73.5	0.250	7.142
r2-preserve	2	0.005	0.001	100.0	0.099	11.525

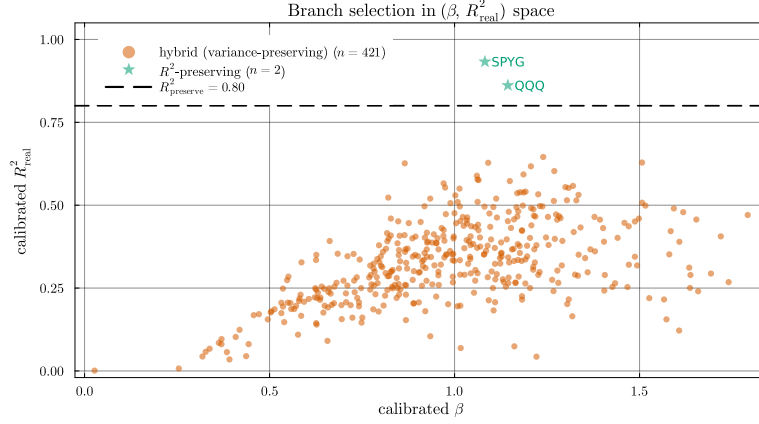


Figure S6: **Branch assignment in calibrated $(\beta, R^2_{\text{real}})$ space.** The $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) separates tracker assets from single stocks. In this universe, only QQQ and SPYG exceed the threshold, and both are assigned to the R^2 -preserving branch. The remaining 421 tickers populate the variance-preserving branch; the clipped variance-preserving branch is unused because the market loading ratio $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$ stays below 0.90 for every ticker.

Table S5: **Aggregate metrics by β quartile, across composition methods.** Quartile cutoffs are computed from the calibrated β distribution across the 423-ticker universe. The hybrid KS pass rate declines from 87.6% in Q1 to 65.9% in Q4 as the market loading ratio ρ_i grows; the naive KS pass rate falls from 24.6% to 1.3% over the same range.

β bucket	Method	$ \hat{\beta} - \beta $	KS pass (%)	W_1	κ	$\text{Var}(g)$
Q1 (low β)	Naive	0.018	24.6	0.403	8.971	18.1
	Gaussian SIM	0.015	0.0	0.620	0.622	14.0
	Hybrid	0.016	87.6	0.177	8.369	14.6
	JumpHMM-on-residuals	0.015	73.2	0.204	7.059	14.3
	Block bootstrap	0.015	71.3	0.208	6.286	13.9
	GARCH(1,1)- t	0.015	57.8	0.253	5.255	13.4
Q2	Naive	0.021	1.9	0.621	7.510	25.1
	Gaussian SIM	0.016	0.2	0.653	1.509	18.0
	Hybrid	0.017	74.9	0.231	6.889	19.0
	JumpHMM-on-residuals	0.017	82.7	0.200	7.799	18.7
	Block bootstrap	0.016	79.8	0.197	7.213	18.0
	GARCH(1,1)- t	0.015	76.4	0.227	6.470	17.8
Q3	Naive	0.023	0.7	0.792	7.390	33.3
	Gaussian SIM	0.017	1.4	0.688	1.947	23.4
	Hybrid	0.018	66.1	0.279	6.904	24.4
	JumpHMM-on-residuals	0.018	78.3	0.242	7.454	24.0
	Block bootstrap	0.017	75.0	0.236	6.888	22.8
	GARCH(1,1)- t	0.016	71.0	0.263	6.265	21.9
Q4 (high β)	Naive	0.029	1.3	0.965	7.233	51.2
	Gaussian SIM	0.022	0.6	0.825	2.054	36.3
	Hybrid	0.022	65.9	0.339	6.739	37.1
	JumpHMM-on-residuals	0.022	69.1	0.313	6.842	37.0
	Block bootstrap	0.021	67.2	0.311	6.334	35.3
	GARCH(1,1)- t	0.021	64.5	0.361	6.064	34.7

Table S6: **Clipping-branch stress test under market-volatility scaling.** The market growth-rate path was rescaled by $\gamma \in \{1, 2, 3\}$ with the calibrated marginals held fixed, and the full hybrid composition method re-evaluated on the 423-ticker universe. As γ rises, the share of clipped tickers grows.

γ	Clipped (%)
1	0.0
2	76.4
3	95.2

Table S7: **Cross-asset generalization of HMM-WJ across three single-stock tickers spanning sector and volatility regimes.** Per-ticker tune outputs (ϵ^*, λ^*) at $n_{\text{paths}} = 200$ and seed = 1234, and the corresponding HMM-WJ pass rates on 1,000 simulated paths at $\alpha = 0.05$ against each ticker’s in-sample (2014–2024) and out-of-sample (2025) growth-rate history. Standard errors in parentheses (binomial SE for pass rates). The (ϵ^*, λ^*) values are reported as the parameter values used for the IS / OoS metrics rather than as canonical per-ticker optima.

Ticker	ϵ^*	λ^*	IS KS (%)	IS AD (%)	OoS KS (%)	OoS AD (%)
NVDA (semiconductors)	10^{-4}	70	98.9 (0.3)	96.7 (0.6)	97.7 (0.5)	96.1 (0.6)
JNJ (health care)	10^{-4}	70	99.4 (0.2)	96.3 (0.6)	74.0 (1.4)	61.3 (1.5)
JPM (financials)	10^{-4}	80	99.1 (0.3)	95.7 (0.6)	79.2 (1.3)	77.7 (1.3)

S7. Cross-asset generalization

We repeated the fitting and evaluation procedure on three stocks: NVDA (semiconductors, high volatility), JNJ (health care, low volatility), and JPM (financials, moderate volatility). Per-ticker tune outputs at $(n_{\text{paths}} = 200, \text{seed} = 1234)$ are reported in Supplementary Table S7. The objective combines ACF and kurtosis errors with fixed weights. Because kurtosis varies across finite samples of heavy-tailed paths, the selected grid point can change with the Monte Carlo sample. The reported (ϵ^*, λ^*) values are reproducible with the stated path count and seed, but they are not unique per-ticker optima. At the selected parameter values, in-sample HMM-WJ KS pass rates were 98.9% for NVDA, 99.4% for JNJ, and 99.1% for JPM. Out-of-sample KS pass rates on the 249-day 2025 hold-out were 97.7%, 74.0%, and 79.2%, respectively.

S8. SIM composition derivations and properties verification

The variance-corrected single-index composition states two closed-form results, the variance scale of Eq. (8) and the residual-variance target of Eq. (11). The derivation below gives the large- T OLS coefficients and composed variance. To derive Eq. (8), first center the full-return draw as in Eq. (6), then set $\varepsilon_i = s_i \tilde{g}_i^c$ for $s_i \in (0, 1]$ and require that the composed series have the same variance as the generator path:

$$\text{Var}(g_i) = \text{Var}(\alpha_i + \beta_i g_m + \varepsilon_i) = \text{Var}(\beta_i g_m + \varepsilon_i) = \sigma_{\text{gen},i}^2, \quad (\text{S18})$$

where α_i drops out because it is constant. Expanding the sum gives:

$$\begin{aligned} \text{Var}(\beta_i g_m + \varepsilon_i) &= \beta_i^2 \text{Var}(g_m) + 2\beta_i \text{Cov}(g_m, \varepsilon_i) \\ &\quad + \text{Var}(\varepsilon_i). \end{aligned} \quad (\text{S19})$$

Substituting $\varepsilon_i = s_i \tilde{g}_i^c$ gives the two moments:

$$\begin{aligned} \text{Var}(\varepsilon_i) &= s_i^2 \sigma_{\text{gen},i}^2, \\ \text{Cov}(g_m, \varepsilon_i) &= s_i \text{Cov}(g_m, \tilde{g}_i^c). \end{aligned}$$

Under the assumption $\text{Cov}(\tilde{g}_i^c, g_m) \approx 0$, the cross term vanishes and the variance condition becomes $\beta_i^2 \sigma_m^2 + s_i^2 \sigma_{\text{gen},i}^2 = \sigma_{\text{gen},i}^2$. Solving for s_i^2 gives Eq. (8).

To derive Eq. (11), write the population R^2 identity for Eq. (4):

$$R^2 = \frac{\beta_i^2 \sigma_m^2}{\beta_i^2 \sigma_m^2 + \sigma_{\varepsilon,i}^2}. \quad (\text{S20})$$

Solving this identity for the residual variance gives Eq. (11). Because $\bar{\varepsilon}_i = 0$ after centering and scaling, the sample mean of a composed path obeys $\bar{g}_i = \alpha_i + \beta_i^{\text{eff}} \bar{g}_m$. Hence, the finite-sample OLS intercept was:

$$\hat{\alpha}_i = \bar{g}_i - \hat{\beta}_i \bar{g}_m = \alpha_i - (\hat{\beta}_i - \beta_i^{\text{eff}}) \bar{g}_m. \quad (\text{S21})$$

Under the zero-cross-covariance assumption, in the large- T limit OLS therefore gives $\hat{\alpha}_i \rightarrow \alpha_i$ and $\hat{\beta}_i \rightarrow \beta_i^{\text{eff}}$. Here $\beta_i^{\text{eff}} = \beta_i$ unless Eq. (10) clips the market loading. On the variance-preserving branch, the composed variance is $\sigma_{\text{gen},i}^2$. Without clipping, $\text{Var}(g_i) = \beta_i^2 \sigma_m^2 + s_i^2 \sigma_{\text{gen},i}^2 = \sigma_{\text{gen},i}^2$. Under clipping, Eq. (10) sets β_i^{eff} to give the same variance. On the R^2 -preserving branch, the composed variance is $\beta_i^2 \sigma_m^2 / R_{i,\text{real}}^2$, the value implied by the calibrated $(\beta_i, R_{i,\text{real}}^2)$ rather than by the generator marginal. The floor guarantees only that $\text{Var}(\varepsilon_i) \geq f \sigma_{\text{gen},i}^2$; it does not guarantee that adding the market term preserves the generator's tail distribution. In Supplementary Table S5, the KS pass rate fell from 87.6% in the lowest-beta quartile to 65.9% in the highest-beta quartile.

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